STAT 17 Discussion Notes
Week 10

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1 Comparing Two Groups

When comparing two groups, the method depends on the type of response variable and whether the samples are independent or paired.

	Two Proportions	Two Means (Independent Samples)	Two Means (Dependent)
Parameter	$p_1 - p_2$	$\mu_1 - \mu_2$	μ_d
Response Variable	Qualitative	Quantitative	Quantitative
Sampling Structure	Independent samples	Independent samples	Matched / paired data
Distribution	Standard Normal (z)	t distribution	t distribution ($df = n - 1$)

2 Two-Proportion Inference

Conditions

- Independent random samples
- Sample size less than 5% of population
- Success-failure condition

$$n_1\hat{p}_1(1-\hat{p}_1) \geq 10$$

$$n_2\hat{p}_2(1-\hat{p}_2) \geq 10$$

Confidence Interval

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Test Statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

3 Two-Sample t Procedures

Conditions

- Independent random samples
- Sample size less than 5% of population
- Nearly normal populations OR large samples ($n \geq 30$)

Confidence Interval

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Test Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

4 Paired t Procedures

Used when observations are paired (e.g., before–after measurements or matched subjects).

Let

$$d = x_1 - x_2$$

where

- d = difference for each pair
- x_1 = first observation in the pair
- x_2 = second observation in the pair

We then treat the differences as a single sample.

Notation

- $\bar{d}$ = sample mean of the differences
- s_d = sample standard deviation of the differences
- n = number of pairs

Confidence Interval

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}}$$

Test Statistic

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$$

5 Chi-Square Test of Independence

Used to determine whether two categorical variables are associated.

Hypotheses

$$H_0 : \text{The two variables are independent}$$

$$H_1 : \text{The two variables are dependent}$$

Conditions

- All expected counts ≥ 1
- No more than 20% of expected counts < 5

Expected Counts

$$E = \frac{(\text{row total})(\text{column total})}{n}$$

Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degrees of Freedom

$$df = (r - 1)(c - 1)$$

P-value

$$p = P(\chi^2 > \chi_0^2)$$

6 Linear Regression

Linear regression describes the relationship between two quantitative variables.

- Explanatory variable: x
- Response variable: y

Regression Line

$$y = mx + b$$

where

- m = slope
- b = y-intercept

For prediction we use

$$\hat{y} = mx + b$$

where $\hat{y}$ is the predicted value of y.

Interpretation of the Slope

The slope m represents the expected change in the response variable y for a one-unit increase in the explanatory variable x.

In context:

> For each one-unit increase in x, the predicted value of y changes by m units on average.

Interpretation of the Intercept

The intercept b represents the predicted value of y when $x = 0$.

> When $x = 0$, the predicted value of the response variable is b.

Note: Sometimes $x = 0$ may not be meaningful in the context of the data.

Residual

A residual measures the difference between the observed value and the predicted value.

$$\text{Residual} = y - \hat{y}$$

where

- y = observed value
- $\hat{y}$ = predicted value from the regression line

Residuals are used to evaluate how well the regression line fits the data.